

Post-fix re-run (17-Jul) — P1–P6 on engine dev 61e8e89

Full re-run of the frozen 35-case engine-vs-reality suite plus every standing suite, against the engine dev stack (`~/fc-estimate-dev/backend-node` on the LND-Dev box, port 4200 / `fc-estimate-dev.figittallabs.com`, verified running commit 61e8e89 : P1 with-package quote, P2 negation guard, P3 named drugs, P4 catch-all guard, P5 newborn question, P6 LOS-banded residuals). Scoring convention as frozen, upgraded exactly as P1 intends: **engine quote per case = the route's number** — package route (`billing_identification = package`) with an un-blocked `package_offer.quote` $\Rightarrow$ `quote.with_package_total`; otherwise `final_estimate` (itemized). New verdict class **QUESTION_RAISED**: flow2 now stops at a mandatory question the FC document itself cannot answer — per the register this is the *intended terminal state*, converted from silent misses.

(a) New scoreboard vs old

Verdict	16-Jul (pre-fix)	17-Jul (post-fix)
ENGINE_GOOD	12	13
ENGINE_OFF	12	9
QUESTION_RAISED	—	3 (BANDANADAM, KRISHNA, G NAGAVENI — converted from silent –74% / +371% / –36% misses)
COURSE_CHANGED	11	10 (AKARAPU re-classified GOOD, justification below)

Accuracy on the comparable clean set — cases that produce a number without needing an unanswerable question (22 post-fix vs 24 pre-fix: the 3 P4 cases became questions, AKARAPU joined from course-changed):

Metric	Engine 16-Jul (n=24)	Engine 17-Jul (n=22)	FC human (same 22)
Median abs % err vs bill (excl. F&B)	26.4%	24.4%	13.6%
Mean abs % err	70.5%	38.8%	40.4%
Within $\pm 25\%$	12/24	11/22	16/22
Engine err $\leq$ human err	9/24	10/22	—
ENGINE_GOOD-case median	14.5%	15.1%	—

Honest bottom line. The catastrophic tail is gone — the mean error halved (70.5% $\rightarrow$ 38.8%, now *better than the humans'* 40.4% on the same cases) because the +371/+421/–87/+82 blowups were fixed or converted to questions. The median barely moved (26.4% $\rightarrow$ 24.4%) for two honest reasons: (i) the neonates swapped one systematic error for another (adult-cohort +79...+143% overquote $\rightarrow$ routine-newborn –73...–83% underquote — the P10 session-load gap plus an unanswerable pathway ambiguity), and (ii) P1's quote headline *exposed* two package-route errors that were previously invisible behind the itemized figure (P7 master drift on GUDURU, P11 extras contamination on PINNAMANENI). Where the extras history is clean, the package quote is spectacular: SURLA –0.9%, AKARAPU –1.8%.

(b) The three most notable swings

- SURLA RAJESWARI: +81.5% $\rightarrow$ –0.9%** (P1+P2 together). The negation guard kept the ₹230k robotic add-on out of "TKR NON ROBOTIC B/L" (itemized 737,751 $\rightarrow$ 507,751), and the package-route quote finished the arithmetic the engine always knew: pkg ORT5531 ₹224,600 + predicted extras ₹178,260 = **₹402,860 vs bill ₹406,505**.
- SHOURYA SRIVASTAV: –86.9% $\rightarrow$ –11.7%** (P3). "INJ STELMA 90 MG" fuzzy-matched to STELARA 90MG/1ML (dose corroborated), ₹104,267 priced from the pharmacy master (sale_rate, flagged), pharmacy replaced-not-added, confirm-drug warning attached. Quote ₹120,586 vs bill ₹136,548.
- Baby of NIKHAT SHAMSI: +18.6% GOOD $\rightarrow$ –83.3% OFF** — the one verdict that *worsened*. P5 now stops at the newborn-pathway question; the FC document supports only "routine newborn care" (no jaundice/NICU wording anywhere in the four neonate FCs), and the routine-newborn build quotes ₹6,109 against a ₹36,605 bill. The old GOOD was the adult cohort's $\sim 2\times$ overquote accidentally landing on a baby that consumed more than routine care. The new number's root cause is P10 (session-based families lose their per-day load: builds ₹6.1–7k vs the routine cohort's own P50 ₹15,315) — and even the cohort P50 sits –50...–58% under these four bills.

(c) Per-case delta table (all 35)

Old err $\rightarrow$ new err at the route's headline number; (R) = recounselling; * = verdict changed.

Case	Old $\rightarrow$ New verdict	Old %	New %	Note
ABDUL HASEEB MOHAMMAD	GOOD $\rightarrow$ GOOD	–13.9	–13.9	unchanged
ARTH AGGARWAL	GOOD $\rightarrow$ GOOD	–25.9	–25.9	unchanged (replay@3 –14%)
Baby of NIKHAT SHAMSI	GOOD $\rightarrow$ OFF*	+18.6	–83.3	see swing 3; replay@2 = 7,012 (–80.8%)
GUDURU LAKSHMI DEVI	GOOD $\rightarrow$ GOOD	–6.4	–31.5	verdict survives only on the frozen in-band rule (bill 348,954 inside itemized band 264k–353k); the new headline quote ₹239,030 is –31.5%: pkg master 149,900 vs billed 168,700 (P7 drift) + extras rung ₹89k vs $\sim$ ₹180k actual implant excludes. Itemized was –6.4%
HEMALATHA THOTA	GOOD $\rightarrow$ GOOD	–3.9	+12.7	package call still exact (ORT5535 @355,000); quote adds ₹61k predicted extras vs ₹14k actual (P11) — error grew but GOOD

Case	Old → New verdict	Old %	New %	Note
K CHARVIN (R)	GOOD → GOOD	+11.1	+11.1	unchanged
LAKSHMI DEVAMMA (R)	GOOD → GOOD	+11.6	−19.3	now scored on the package quote ₹550,860 (pkg 142,600 room-tier + extras incl. the validated robotic ₹230k); bill is a package bill (ORT5531)
Neeharika Gorla	GOOD → GOOD	−15.1	−15.1	unchanged
PINNAMANENI VIJAYALAKSHMI	GOOD → OFF*	−1.5	+29.8	package call still exact (ORT5536 @690,000 = the bill's package) but the quote adds ₹218,847 predicted extras vs ₹10,340 actual — P11 leak the quote band gate did not catch . Replay@actual-LOS quote +28.5% ⇒ not a LOS effect
RAVULA PUSHYAMI	GOOD → GOOD	−21.8	−21.8	unchanged
S RAMESH (R)	GOOD → GOOD	−15.5	−15.5	unchanged
SHIVA KUMARI	GOOD → GOOD	+22.8	+22.8	unchanged (hand-kept as 16-Jul; replay@2 +8%)
BANDANADAM NAVEEN IGNEIOUS	OFF → QUESTION*	−73.8	—	P4 stop; both options offered are generic (general_plastic, general_surgical) — the FC wording names no onboarded procedure; no non-generic what-if exists
Baby Boy NAKKA TRIVED RAJ K.	OFF → OFF	+106.5	−77.1	routine-newborn answer per FC doc; sign flip (P10)
Baby of HARIKA	OFF → OFF	+79.1	−80.1	same
Baby of SOUNDARYA	OFF → OFF	+142.7	−73.0	same
G NAGAVENI	OFF → QUESTION*	−35.8	—	P4 stop; options: generic ortho + wound_debridement_and_soft_tissue_repair — the latter is NOT an obvious match for an Achilles tendon rupture repair, no what-if forced; the ₹27,500 TENDON pkg stays correctly blocked (not_ready)
KRISHNA B R	OFF → QUESTION*	+371.1	—	P4 stop; options: generic plastic / generic ortho / emergency-trauma — no maxillofacial-ish family exists in the registry (recorded honestly, nothing forced)
NARESH GURAJALA	OFF → OFF	+420.7	+217.6	P6 partial win: 41,068 → 25,048 (Investigations 20,790 → 4,770 via the 90-case short-stay sub-cohort); the residual is the PF/room/per-day machinery P6 deliberately scoped out
RAMULU GUNDI	OFF → OFF	+27.0	+27.0	unchanged watch-list marginal
SHOURYA SRIVASTAV	OFF → GOOD*	−86.9	−11.7	P3 (see swing 2)
SURLA RAJESWARI	OFF → GOOD*	+81.5	−0.9	P1+P2 (see swing 1)
VARSHITHA	OFF → OFF	−64.4	−29.1	frozen wording names no drug ⇒ P3 guard path correct ("confirm the drug" warning fired, no silent augmentation). The numeric improvement comes from family re-resolution (general_medical_management_infusion → chemotherapy_systemic_therapy_infusion_daycare under the updated registry), not from a drug line. Post-confirm figure (drug confirmed by the FC): −21.8% per the register's verified run
Y CHANDRIKA	OFF → OFF	−33.3	−33.3	unchanged (bilateral EVLT, deferred)
A JIVITESH SAI (R)	COURSE → COURSE	+13.4	+13.4	unchanged
AKARAPU VISHWANATAHM	COURSE → GOOD*	+34.3	−1.8	re-classified with justification: the bill IS a package bill (ORT5510), so the LOS 4→2 change is insulated by the package — the P1 quote at <i>planned</i> inputs (₹239,030) lands −1.8% on its own merits
ANITA KUMARI (R)	COURSE → COURSE	−73.4	−73.4	route = non_package, so the ₹77,000 GYN5219 offer is correctly NOT the headline (its own conversion warning fired)
HARANADHA BABU	COURSE → COURSE	−80.2	−80.2	unchanged
K ASHWINI	COURSE → COURSE	−12.8	−12.8	unchanged (replay@4 −9%)
KRANTHI KUMAR	COURSE → COURSE	−74.8	−74.8	unchanged (FB extraction inconsistent)
MEENAKSHI VECHA	COURSE → COURSE	−8.0	−8.0	unchanged. NOTE: family pinned to adenotonsillectomy (the engine's own 16-Jul resolution) via the UI's <code>selections.family</code> mechanism — persistent Gemini stray-brace matcher flake on this wording (see (f)3)
RAMBABU (R)	COURSE → COURSE	−67.8	−67.8	unchanged (urology combo, P8)
SARBESWAR MAITY	COURSE → COURSE	−86.2	−86.2	unchanged
VUTKUR SURYA PRAKASH (R)	COURSE → COURSE	−31.9	−31.9	route = non_package ⇒ itemized stays headline; an ENT5147 quote (₹135,802, −11.6%) exists on the response but is correctly not the route's number
YALABANDI K S L VALLIDEVI	COURSE → COURSE	+14.6	+14.6	unchanged

Newborn-question answers documented per case (answered from the FC document ONLY, never the bill): NAKKA / HARIKA / SOUNDARYA / NIKHAT all → routine_newborn_care_and_vaccination , basis "FC doc describes routine newborn medical management (no jaundice/NICU wording)". None of the four neonate FC documents mentions phototherapy, jaundice or NICU.

(d) Standing suite results

Suite	Baseline	Post-fix (17-Jul)	Verdict
flow2 validation sweep	374 cases, 6 failures (5x I4 family-flip + 1 transient HTTP), matcher-flake 46	374 cases (619 evals, 954s), 1 failure, matcher-flake 23, questions 230/230 answered, no-match 0	the I4 family-flip class is GONE (matcher cache pin); the single failure is a stale suite pin, not a regression — see (f)2
Cash parity (TKR pinned to the paisa)	1042/1042	PASS=1042 FAIL=0 (validate_estimate) + validate_artifacts clean (26/26 cohort rows, 53/53 service rows)	✅ pin intact
Workbook cell-parity (validate_workbook)	(part of npm test)	ENV-BLOCKED	crashes at file-read before any engine code: its reference extraction (full_cell_data.json + parity_spec.json) is hard-coded to an old session scratchpad path that no longer exists on any machine (checked box + Mac). Not a P1–P6 regression; the fix is to commit the reference extraction into spec/ like sheet_targets.json
Normal suite	52/52 (pre-onboarding registry)	PASS=526 FAIL=0 , 130 data flags	✅ zero failures; the check count grew 52 → 526 because the registry now carries 170 families × 3 rooms + insurer/package orgs + workbook smokes; the 130 flags are the known DATA_CONCERNS classes (template-less Investigations/Pharmacy backfilled from historical actuals)
Insurance edge suite	32 cases, 0 bugs	32 cases, 0 bugs, 0 warnings	✅
verify-estimates harness (207 zero-input builds)	15-Jul first run 25/182; after the 15-Jul Q1/Q3/Q4 fixes 32 in-band / 175 out (classes: 54 token-OT, 10 investigations-0, 1 pharmacy-0, 15 gross, 3 conversion)	207 builds, 0 crashes, 32 in-band / 175 out — class counts identical (54/10/1/15/3)	P1/P3/P6 moved <i>nothing</i> here, and that is the designed outcome, not a miss: the harness runs zero-input builds — no treatment text so P3 cannot fire, default LOS = cohort P50 so P6's ≤P25 banding cannot trigger, and P1's quote is additive and never touches buckets. The movement the fixes deliver appears exactly where the inputs exist — proven in the 35-case replay (SHOURYA –87→–12, NARESH +421→+218, SURLA +82→–1)

flow2 sweep composition shift vs 16-Jul (informational): reached-numbers 327→351, package path 69→55, non-package 258→296, questions 215→230 — the added questions are P4/P5's (the suite answers them with the majority option, as a user would).

(e) Remaining misses + causes (9 OFF)

Case(s)	New err	Cause
4 neonates (NAKKA –77, HARIKA –80, SOUNDARYA –73, NIKHAT –83)	–73... –83%	P10 (session-based newborn families lose their per-day load: builds ₹6.1–7k vs the routine cohort's own P50 ₹15,315) + a genuine intake ambiguity: even the cohort P50 sits ~50...–58% under these bills — these babies plausibly consumed jaundice/NICU-level care the FC documents never recorded. The P5 question is the right machinery; the FC must answer it from the chart
PINNAMANENI	+29.8%	P11 : extras rung contaminated by the package's own billed lines (₹218,847 predicted vs ₹10,340 actual on cash robotic-TKR bilateral); the quote band gate missed this one — tighten it with the billed-total-range check the conversion warning already computes
NARESH	+217.6%	P6 residual (deliberately scoped out): PF/room/per-day machinery still stay-independent; small absolute money
VARSHITHA	–29.1%	P3 guard path working as designed (wording names no drug); –21.8% once the FC confirms the drug
Y CHANDRIKA	–33.3%	bilateral EVLT laterality/consumables (watch-list, deferred)
RAMULU	+27.0%	marginal Investigations inflation (watch-list)

Near-miss inside GOOD worth tracking: **GUDURU's headline quote is –31.5%** (P7 package-master drift ₹149,900 vs billed ₹168,700 + extras ₹89k vs ~₹180k actual implant excludes) — the verdict survives only on the frozen in-band rule. Same quote (₹239,030) was –1.8% on AKARAPU: one number cannot fit both patients because the implant load (the real money in GIPSA TKR) varies per case — the P7 sync audit plus implant-aware extras are the fix.

(f) Expectation-vs-regression flags

- P1 exposes P7/P11 on package quotes — real behavioral regression on 2 cases, flagged loudly.** PINNAMANENI (GOOD→OFF, +29.8%) and GUDURU (headline –31.5%, GOOD only via band): the with-package quote is now the headline on the package route, so stale master amounts (P7) and contaminated extras history (P11) hit the quoted number where they were previously invisible behind the itemized figure. Both are already-queued register entries; this re-run puts case-level rupee numbers on them. Suggested tightening: block the quote when its with-package total falls outside the package's own billed-total range (the conversion check already computes exactly that range — SURLA's fired while its quote was –0.9%, so the gate must compare the QUOTE, not the converted itemized total).
- flow2 sweep's single failure is a stale pin — expectation needs updating, and it predates P1–P6.** The `mode-both: numbers.note missing` probe asserts a placeholder that existed at suite-commit time (`numbers.note = 'logic comparison lands in phase B'`); phase B landed the same day (16-Jul) and replaced it with the real `numbers.logic + numbers.comparison` payload — both present in the probe's response. The suite was NOT edited; flagged here for the owner to update the pin.
- Gemini matcher stray-brace flake (infrastructure, new register candidate P13).** `gemini-3.1-pro-preview` intermittently returns valid JSON followed by a stray `}`; `geminiJson`'s fallback regex (`/\{[\s\S]*\}/`, greedy to the LAST brace) then fails too, so the whole 3-try retry ladder surfaces "AI family matcher unavailable". Root-caused live on the box (raw model text ends `}\n`). During the replay, "TOTAL KNEE REPLACEMENT (TKR) - LEFT" recovered after ~4 API calls (then the 10-min wording cache held); MEENAKSHI's misspelled coblation wording did NOT recover in 12+ calls and was pinned via `selections.family` (documented in the table). The sweep saw 23 such flakes (all retried internally). One-line fix in `src/modules/ai/gemini.js` (balanced-brace extraction) — NOT applied; engine untouched.
- Normal-suite count 52 → 526 and verify-harness baseline 25/182 vs 32/175:** both are registry/fix evolution between the pinned wordings of the task brief and the current dev state, not regressions — 0 failures in the normal suite either way; the harness matches the 15-Jul *post-Q-fix* baseline exactly.

5. **validate_workbook ENV-BLOCKED** (see suite table) — reference extraction files were in a session scratchpad that tmp-cleanup removed; never existed on the box. Recommend committing them under `spec/`.
6. **No pinned suite was edited; nothing was committed or pushed.**

Method notes for this re-run

- Runner: `run_engine_postfix.mjs` — a copy of the frozen `run_engine.mjs` with exactly three deltas: (i) sends `patient: { name }` on flow2 (the baseline never did; it powers P5's newborn detection), (ii) answers the newborn question only from the FC document (mapping: jaundice/phototherapy wording → jaundice option, NICU wording → NICU option, else routine — all four hit "else routine"), (iii) stops and records the P4 catch-all question instead of auto-answering it. `COURSE_CHANGED` replays at actual LOS/ward re-run identically.
- Suites ran sequentially on the box (they share the engine + DB): flow2 sweep → normal → edge → `npm test` (artifacts + 1042 parity + workbook) → verify-estimates.
- Artifacts: `/private/tmp/fc-eval/engine_runs_postfix.jsonl`, `comparison_postfix.jsonl`, `adjudicated_postfix.json`, `run_engine_postfix.mjs`, `compare_postfix.py`; box logs `/tmp/flow2_sweep.log`, `/tmp/normal_suite.log`, `/tmp/edge_suite.log`, `/tmp/parity.log`, `/tmp/verify_estimates.log`. All 16-Jul baseline artifacts untouched.

Engine vs Reality — 35 historical FC counsellings vs 162 final bills (16-Jul)

Real-world validation of the FC estimate engine (<https://fc-estimate-dev.figitallabs.com>) against KIMS Gachibowli's own paperwork: 35 historical Financial Counselling PDFs (`~/Downloads/inputs/Old Financial Counsellings/`) and their matching final bills (`~/Downloads/inputs/Old Final Bills/` , 162 files; all 35 FCs matched an FB by name and IP number, the other 127 FBs have no FC and were ignored).

Method in one paragraph. Each FC and FB PDF was extracted to structured JSON with Gemini (`gemini-2.5-flash` , temperature 0, JSON mode); 4 extractions (2 FC, 2 FB) were hand-verified against the raw PDFs before running the fleet. Per case the engine was driven exactly as a user would: free-text treatment (surgery name + the truer counsellor-remarks wording) through `P0ST /api/flow2/evaluate` for family + billing-route resolution (answering pending questions from the FC's own fields: care type, setting, robotic), then `P0ST /api/estimate/build` with the FC's opted room and planned LOS (`los_basis: Manual`), plus a default-LOS build and — where the actual LOS differed — **a replay at the actual LOS/ward**. Insurers were resolved via `/api/lookup/organizations` from the bill's Organization (preferred) or the FC's TPA (e.g. New India/Oriental/National/United → GIPSA Insurance; Star/Care/Manipal/MD India → Non-GIPSA). All bill comparisons use **net amount (after concession) minus Food & Beverages**, since engine totals exclude F&B; gross-vs-net is flagged where it matters. Every number below is reproducible from the run artifacts (paths at the end).

Two verdict rules follow the caution in the brief: bills whose treatment changed (LOS jump, added procedures, multi-procedure "second procedure at 50%/25%" combos) are scored **COURSE_CHANGED**, never as engine misses; and package-billed cases are judged on the engine's own chosen billing route (its package identification), not only the itemized number.

(a) Headline scoreboard

Verdict	Cases	Meaning
ENGINE_GOOD	12	same treatment/LOS; engine quote within $\pm 25\%$ of bill-excl-F&B (or exact package call)
ENGINE_OFF	12	same treatment/LOS; engine still off — real engine misses, root-caused below
COURSE_CHANGED	11	LOS/procedures/billing structure changed vs the FC plan — scored separately
NO_MATCH	0	the deployed registry (30+ families incl. catch-alls) matched everything — but 3 of those matches were <i>wrong</i> , counted in ENGINE_OFF

Accuracy on the 24 clean (non-course-changed) cases:

Metric	Engine	FC human (same 24 cases)
Median abs % error vs bill (excl. F&B)	26.4%	12.0%
Mean abs % error	70.5%	40.7%
Cases within $\pm 25\%$	12/24	15/24
Engine error $\leq$ human error	9/24	—

Honest bottom line: the engine is not yet as accurate as the human FCs overall (26% vs 12% median), but its misses are concentrated in 5 specific, fixable causes (section d) — and on structured surgical families it already matches or beats the humans. On the 12 ENGINE_GOOD cases the median error is 14.5%, and on package identification it was *exact* (code and rupee amount) on both cash robotic-TKR package bills.

Calibration context — the humans are not oracles either: the FC's quoted **range contained the actual bill in only 11/35 cases** (bill below the range 13x, above it 11x). Median human error across all 35 billed cases: 14.1%. The engine quote landed inside the FC's own range 9/35 times, below it 18x, above 8x.

(b) Full per-case table

Engine quote = the engine's number on its own chosen billing route (package amount where its route was package *and* that is what it would quote; itemized `final_estimate` otherwise), at the FC's planned LOS and opted room. Bill = net minus F&B. LOS = planned/actual. Eng% / FC% = engine and FC-range-midpoint deviation from the bill. (R) = Recounselling (mid-treatment estimate).

Patient	Treatment (FC)	Payor	FC range	Engine quote (route)	Bill xF&B	LOS p/a	Eng%	FC%	Verdict	Cause / note
Mr ABDUL HASEEB MOHAMMAD	LAP./VATS THORACIC SYMPATHECTOMY	Cash	262,000–321,000	240,424 (non-pkg)	279,340	2/2	-14%	+4%	ENGINE_GOOD	thoracic_surgery_vats; bill gross was 329,640, hospital gave ₹50,300 concession
Master ARTH AGGARWAL	MEDICAL MANAGEMENT	GIPSA	72,000–88,000	82,764 (non-pkg)	111,619	2/3	-26%	-28%	ENGINE_GOOD	stayed 1 day longer; replay@3 = 95,935 (-14%)
Baby of NIKHAT SHAMSI	MEDICAL MANAGEMENT (newborn)	Cash	41,000–50,000	43,417 (non-pkg)	36,605	1/2	+19%	+24%	ENGINE_GOOD	within 25% despite LOS 1→2
Mrs GUDURU LAKSHMI DEVI	TKR - LEFT (conventional)	GIPSA	399,000–439,000	326,703 (pkg route)	348,954	4/4	-6%	+20%	ENGINE_GOOD	bill = pkg ORT5510 168,700 + implant excludes; engine itemized -6%
Mrs HEMALATHA THOTA	ROBOTIC TKR - UNILATERAL - RIGHT	Cash	355,000–391,000	355,000 (package)	369,220	3/3	-4%	+1%	ENGINE_GOOD	package call exact: ORT5535 @ 355,000 = the bill's package
Master K CHARVIN (R)	ORCHIDOPEXY	GIPSA	161,000–196,000	222,884 (non-pkg)	200,680	1/1	+11%	-11%	ENGINE_GOOD	paediatric_laparoscopic_surgery_major (medium conf) — right money anyway
K LAKSHMI DEVAMMA (R)	TKR BILATERAL (robotic)	GIPSA	609,000–670,000	762,181 (non-pkg run)	682,744	6/5	+12%	-6%	ENGINE_GOOD	engine's robotic add-on ₹230,000 appears verbatim on the bill (ROBO (TKR) - BILATERAL)
Mrs Neeharika Gorla	MEDICAL MANAGEMENT	GIPSA	70,000–85,000	82,764 (non-pkg)	97,488	2/1	-15%	-20%	ENGINE_GOOD	
Mrs PINNAMANENI VIJAYALAKSHMI	ROBOTIC TKR - BILATERAL	Cash	690,000–760,000	690,000 (package)	700,340	6/4	-2%	+4%	ENGINE_GOOD	package call exact: ORT5536 @ 690,000; LOS 6→4 irrelevant under package
Mrs RAVULA PUSHYAMI	EXCISION SOFT TISSUE LESION - MAJOR	GIPSA	69,000–84,000	61,014 (non-pkg)	78,043	1/1	-22%	-2%	ENGINE_GOOD	
Mr S RAMESH (R)	OTHER MAJOR SURGERY ORTHOPAEDICS	Cash	282,000–345,000	242,160 (non-pkg)	286,709	3/3	-16%	+9%	ENGINE_GOOD	matched clavicle_fracture_fixation off generic wording — landed anyway (lucky; see d.5)
Mrs SHIVA KUMARI	ORIF CLAVICLE PLATING	Cash	295,000–360,000	257,877 (non-pkg)	209,920	4/2	+23%	+56%	ENGINE_GOOD	replay@2 = 226,443 (+8%); human was +56%
Mr BANDANADAM NAVEEN IGNESIOUS	OTHER MAJOR SURGERY PLASTIC SURGERY	Cash	442,000–541,000	118,019 (non-pkg)	449,879	3/2	-74%	+9%	ENGINE_OFF	vague wording → generic plastic family, <i>high</i> confidence
Baby Boy NAKKA TRIVED RAJ K.	MEDICAL MANAGEMENT (newborn)	Cash	65,000–80,000	63,107 (non-pkg)	30,555	2/1	+106%	+137%	ENGINE_OFF	neonatal gap (human was +137%)
Baby of HARIKA	MEDICAL MANAGEMENT (newborn)	Cash	60,000–73,000	63,107 (non-pkg)	35,231	2/3	+79%	+89%	ENGINE_OFF	neonatal gap
Baby of SOUNDARYA	MEDICAL MANAGEMENT (newborn)	Cash	63,000–77,000	63,107 (non-pkg)	26,007	2/1	+143%	+169%	ENGINE_OFF	neonatal gap
G NAGAVENI	ACHILLES TENDON RUPTURE REPAIR	GIPSA	252,000–307,000	164,978 (pkg?)	256,866	3/2	-36%	+9%	ENGINE_OFF	orthopaedic_management_procedure catch-all (medium conf) underprices; billing-id offered TENDON REPAIR pkg @27,500 (wrong scale)
Mr KRISHNA B R	ARCH BAR WIRING – DOUBLE JAW	Cash	30,000–37,000	86,120 (non-pkg)	18,282	1/1	+371%	+83%	ENGINE_OFF	mis-resolved maxillofacial → general_plastic_surgery (medium conf)
Mr NARESH GURAJALA	MEDICAL MANAGEMENT (1 day)	Cash	25,000–30,000	41,068 (non-pkg)	7,887	1/1	+421%	+249%	ENGINE_OFF	medical-mgmt template floor (₹20,790 P50 investigations etc.) swamps a trivial stay
Mr RAMULU GUNDI	CRIF CALCANEUS FRACTURE	Cash	247,000–302,000	328,743 (non-pkg)	258,837	4/3	+27%	+6%	ENGINE_OFF	marginal; Investigations bucket ₹55,490 looks inflated
Mr SHOURYA SRIVASTAV	DAY CARE INJ STELMA 90 MG IV	Cash	133,000–162,000	17,821 (non-pkg)	136,548	1/1	-87%	+8%	ENGINE_OFF	named biologic (₹~1.2L) not priced from treatment text
S RAJESWARI	TKR BILATERAL — "NON ROBOTIC"	GIPSA	393,000–432,000	737,751 (as-run)	406,505	6/5	+82%	+2%	ENGINE_OFF	bug: "NON ROBOTIC" text triggered the ₹230,000 robotic add-on (robotic:auto); clean rebuild 507,751 still +25%; pkg route (224,600 + ~190k implants = 415k) would have been ~+2% but the engine never emits that total
Miss VARSHITHA	injection day care procedure	Cash	51,000–62,000	17,821 (non-pkg)	50,000	1/1	-64%	+13%	ENGINE_OFF	named drug not priced
Mrs Y CHANDRIKA	ENDOVENOUS ABLATION B/L + FOAM SCLERO	Non-GIPSA	132,000–161,000	115,798 (non-pkg run)	173,668	1/1	-33%	-16%	ENGINE_OFF	right family (EVLt, high conf); bilateral + laser consumable underpriced; billing-id suggested pkg VAS0033 @280,000 while the bill was open

Patient	Treatment (FC)	Payor	FC range	Engine quote (route)	Bill xF&B	LOS p/a	Eng%	FC%	Verdict	Cause / note
Master A JIVITESH SAI (R)	ADENOTONSILLECTOMY (coblation)	GIPSA	137,000–168,000	129,354 (non-pkg)	114,097	1/1	+13%	+34%	COURSE_CHANGED	billed as combo: TONSILLECTOMY pkg 40,100 + ADENOIDECTOMY 50% + 2xEUM 25% + COBLATOR ₹16k; engine itemized was still +13% (human +34%)
Mr AKARAPU VISHWANATAHM	TKR - LEFT	GIPSA	264,000–291,000	326,703 (pkg route)	243,307	4/2	+34%	+14%	COURSE_CHANGED	LOS halved; replay@2 = 297,788 (+22%); bill pkg ORT5510 @168,700 vs engine pkg master 149,900 (see data gaps)
Mrs ANITA KUMARI (R)	"d & c procedure" (medical recouns.)	GIPSA	106,000–129,000	38,368 (non-pkg)	144,512	4/4	-73%	-19%	COURSE_CHANGED	bill = D&C package + discounted hysteroscopy combo; treatment text also mis-resolved to lscs_caesarean at low confidence (D&C family not onboarded)
Mr T HARANADHA BABU	PTCA - SINGLE VESSEL	GIPSA	500,000–550,000	201,655 (non-pkg)	1,019,024	3/13	-80%	-48%	COURSE_CHANGED	13-day stay, TWO packages (CAR5154 PTCA + CAR0122 CAG), IVUS, 2nd vessel, dialysis; replay@13 = 426,748 — still -58%, but the treatment genuinely multiplied
Mrs K ASHWINI	POEM - PC	GIPSA	205,000–226,000	218,312 (pkg route)	250,220	1/4	-13%	-14%	COURSE_CHANGED	stayed 4 days not 1; replay@4 = 228,368 (-9%) — engine replay lands
Mr KRANTHI KUMAR	MEDICAL MANAGEMENT (AICU)	Cash	307,000–376,000	102,488 (non-pkg)	406,000	4/1*	-75%	-16%	COURSE_CHANGED	MICU + endoscopic sclerotherapy/EVL added; *FB extraction internally inconsistent (category sum ≠ gross) — LOS suspect
Baby MEENAKSHI VECHA	ADENOTONSILLECTOMY + EUM	GIPSA	127,000–156,000	131,854 (non-pkg)	143,368	1/1	-8%	-1%	COURSE_CHANGED	billed as ADENOIDECTOMY pkg 50,000 + tonsillectomy 50% + 4 lines at 25% + COBLATOR ₹16k; engine itemized still -8%
Mr RAMBABU (R)	CYSTOSCOPY + URS LITHOTRIPSY	GIPSA	421,000–464,000	128,677 (non-pkg)	399,282	2/6	-68%	+11%	COURSE_CHANGED	combo (CYSTOSCOPY URS + DJ STENTING pkg 64,400 + 50% lines) + LOS 2→6; replay@6 = 179,506, still -55% — urology combos are a real gap
Mr SARBESWAR MAITY	medical management	Non-GIPSA	163,000–200,000	68,458 (non-pkg)	495,516	3/16	-86%	-63%	COURSE_CHANGED	LOS 3→16; replay@16 = 199,931, still -60% — long-stay escalation beyond the family cohort
Mr VUTKUR SURYA PRAKASH (R)	ENDOSCOPY	Cash	171,000–208,000	104,605 (non-pkg)	153,554	4/3	-32%	+23%	COURSE_CHANGED	bill added colonoscopy + therapeutic endoscopy + fibroscan + 2D echo; bill sits inside the engine's (very wide) P25–P75 band 47,689–169,790
Mrs YALABANDI K S L VALLIDEVI	LAP MYOMECTOMY	Non-GIPSA (Star)	146,000–161,000	190,560 (non-pkg)	166,213	3/1	+15%	-8%	COURSE_CHANGED	billed as <i>Open</i> Myomectomy pkg 107,502 + HYSTEROSCOPY 50% combo, LOS 3→1; engine itemized still +15%

(c) Where the engine is CORRECT — patterns

- Package identification is the engine's strongest skill.** On both cash robotic-TKR package cases it produced the exact package the hospital billed, to the rupee: HEMALATHA ORT5535 @355,000 (bill 369,220 incl. ₹14k extras, -3.9%) and PINNAMANENI ORT5536 @690,000 (bill 700,340, -1.5%). The flow2 billing-identification step also correctly said "package" on every case the hospital actually billed as a package (8/8), and "non-package" on ABDUL, S RAMESH, SHIVA etc. that were billed open.
- The robotic add-on model is validated by reality.** LAKSHMI DEVAMMA's bill carries the line "ROBO (TKR) - BILATERAL ₹230,000" — exactly the contracted add-on amount and item the engine priced (OTI0099, tariff_contracted). Total +12%.
- GIPSA joint replacement, itemized route:** GUDURU -6%, LAKSHMI +12%, AKARAPU replay +22% — the insurer-tariff itemized totals track pkg+implant-excludes bills well, because implants (the money) are excluded from GIPSA TKR packages and the engine's pharmacy/implant history carries them.
- Clean-wording mid-size surgery, cash or GIPSA:** VATS sympathectomy -14%, orchidopexy +11%, clavicle ORIF +23%→+8% on replay, soft-tissue excision -22%. When the treatment text names a real procedure and a specific family exists, the engine is inside ±25% almost every time.
- Adult inpatient medical management (insured, 1–3 days):** ARTH -26%→-14% replay, NEEHARIKA -15%, Baby-of-NIKHAT +19%. The general_medical_management family is decent at *adult-scale* short stays.
- LOS replay works.** Of the course-changed cases where only the stay length changed, replay at the actual LOS landed within ±25% in 3 of 4 (ASHWINI -9%, AKARAPU +22%, plus ENGINE_GOOD-side replays SHIVA +8%, ARTH -14%). This is direct evidence the driver model is right and the *input* (planned LOS) is what fails — same failure the humans have.
- ENT combos are absorbed by history.** Even though the engine prices a single family, its adenotonsillectomy cohort evidently contains combo bills: JIVITESH +13%, MEENAKSHI -8% vs real pkg+50%+25% combo bills — while the human FC was +34% on JIVITESH.

(d) Where the engine is WRONG — root-caused (12 real misses)

- 1. No "with-package" total on the package route — 2 cases directly, affects every package quote (₹100k–330k per quote).** The engine identifies the package but `final_estimate` stays itemized and `package_offer` carries no payable-extras/with-package total (both null in every response). SURLA: itemized +82% (or +25% clean); the package math it already had (pkg 224,600 + ~190k implant excludes) reproduces the ₹406,505 bill almost exactly. G NAGAVENI's billing-id offered a ₹27,500 TENDON REPAIR package with `not_ready` status while the itemized total was -36% — no usable number either way. *The engine knows the route; it doesn't finish the arithmetic.*
- 2. Robotic add-on false-trigger on negated wording — 1 case, ₹230,000.** Treatment text "TKR **NON** ROBOTIC B/L" with `controls.robotic: "auto"` fires the `treatment_text_robotic` detector (substring match) and injects the ₹230,000 add-on. `robotic: "no"` correctly suppresses it (verified: 737,751 → 507,751). One-line negation guard in the robotic text detector (`services.js isRoboticText`).
- 3. Named high-cost drugs invisible to daycare infusion pricing — 2 cases, ₹32k–119k each.** "DAY CARE INJ STELMA 90 MG" → engine ₹17,821 vs bill ₹136,548 (-87%); VARSHITHA -64%. The drug name is in the treatment text and in the pharmacy master (MRP lookup exists via `/api/lookup/pharmacy-items`), but the infusion family prices only the generic cohort. The human FCs were within 8–13% on both — they looked the drug up.
- 4. Neonatal coverage gap — 3 cases (+ the floor problem), +79% to +143%.** "Baby of ..." cases resolve to adult `general_medical_management` (high confidence) and quote ₹63,107 for what bills at ₹26k–36k. Expected NO_MATCH per the brief; instead the engine confidently overquotes x2. Note: the human FCs were *equally* wrong (+89% to +169%) — the hospital systematically overquotes newborn stays, so history-mining a "newborn" family from the FB corpus would beat both.
- 5. Catch-all families match vague wording with unearned confidence — 3 cases, ₹68k over to ₹332k under.** BANDANADAM ("OTHER MAJOR SURGERY PLASTIC SURGERY") → generic plastic family at *high* confidence, ₹118k vs a ₹450k reconstruction. KRISHNA (ARCH BAR WIRING, maxillofacial) → same plastic family, +371%. G NAGAVENI (Achilles repair) → orthopaedic catch-all, -36%. When the best match is a catch-all and the wording names no specific onboarded procedure, the gate should stop with a question (flow2 already has the pending-question machinery), not quote.
- 6. Trivial-stay floor — 1 case, +421%.** NARESH: 1-day observation billed ₹7,887; the medical-management template's P50-backfilled rows (e.g. Investigations ₹20,790 "filled from historical actuals") produce a ₹41k floor. Small absolute money, terrible optics.
- 7. Laterality/equipment nuance — 1 case, -33%.** Y CHANDRIKA bilateral EVLT + foam: right family, but bilateral consumables (laser fiber x2) and PF underpriced; the payor-history package suggestion (RF ABLATION @280,000) contradicted the open bill.
- 8. Marginal overestimate — 1 case, +27%.** RAMULU CRIF calcaneus; Investigations bucket ₹55,490 alone is bigger than most whole-case investigation spends here.
- Data gaps observed (not scored):** engine package master amounts lag the billed GIPSA amounts — ORT5510 engine 149,900 vs billed 168,700 (Oriental & New India); ORT5531 engine 224,600 vs billed 202,100 (SURLA) and 252,600 (LAKSHMI) — same code, two billed amounts, suggesting tariff-version drift the master doesn't capture. HARANADHA's CAR5154/CAR0122 package codes were not verified against our master. The engine's own "Package conversion check" warning fired on exactly the packages whose converted totals are unreliable (AKARAPU, SURLA, ANITA) — the engine already knows.

(e) Course-changed cases (11) — what actually changed

Case	What changed between FC and bill	Engine replay at actuals
HARANADHA BABU	PTCA became PTCA+CAG (2 packages) + IVUS + 2nd vessel + dialysis; LOS 3→13	426,748 vs 1,019,024 — replay off; treatment multiplied, not just LOS
SARBESWAR MAITY	medical mgmt 3d → 16-day escalation	199,931 vs 495,516 — off; cohort can't scale to outlier stays
RAMBABU (R)	URS became multi-procedure combo (pkg 64,400 + 50% lines); LOS 2→6	179,506 vs 399,282 — off; urology combo gap
K ASHWINI	POEM stay 1→4 days; FC said Package, billed open	228,368 vs 250,220 (-9%, replay good)
AKARAPU	TKR LOS 4→2 (package bill insulated most of it)	297,788 vs 243,307 (+22%, replay acceptable)
KRANTHI KUMAR	AICU medical plan → MICU + EVL/sclerotherapy; FB extraction inconsistent (sum ≠ gross)	replay not meaningful
ANITA KUMARI (R)	medical recouns. → D&C package + hysteroscopy combo	n/a (family mis-resolved anyway)
A JIVITESH SAI (R)	single-family plan billed as 4-procedure ENT combo + coblator	itemized already +13%
MEENAKSHI VECHA	same, 6-procedure ENT combo + coblator	itemized already -8%
YALABANDI	lap myomectomy billed as <i>open</i> pkg + hysteroscopy 50%; LOS 3→1	itemized already +15%
VUTKUR (R)	"endoscopy" became UGI + colonoscopy + therapeutic + fibroscan	bill inside wide band; typical -32%

7 of 35 FCs were **Recounselling** (mid-treatment) — 4 of them ended course-changed. Recounselling inputs need consumed-to-date context; scoring them like initial estimates flatters nobody.

(f) Fix list, ranked by money impact

- Emit the with-package total** (package amount + predicted payable extras/excludes) whenever `billing_identification` says package, and make it the headline quote on that route. Impact here: ₹300k+ error removed on SURLA alone; 8/35 real bills were package bills.
- Negation guard on the robotic text detector** ("NON ROBOTIC", "non-robotic", "without robot"). ₹230,000 per false trigger; trivial fix; also mirror it in any UI default.
- Price named drugs in daycare/infusion treatment text** via the existing pharmacy-item lookup (MRP × qty), added on top of the family cohort. ₹32k–119k per case; daycare infusion volume is high.
- Gate guard for catch-all families:** when the top match is a generic family (`general_plastic`, `orthopaedic_management`, `general_medical...`) and no specific procedure token matched, return a pending question instead of a high-confidence quote. Would have converted 3 big misses (-74%, +371%, -36%) into questions.
- Neonatal family** (Baby-of/newborn detection → nursery/NICU cohort mined from bills like these: ₹26k–36k @1–3 days). Also fixes the medical-mgmt floor for trivial stays (NARESH).
- Refresh per-tariff package amounts** (ORT5510/ORT5531 drift of ₹19k–28k vs billed) and surface the engine's own package-conversion warning as a quote blocker, not a footnote.

7. **Combo quoting** through flow2 multi-path (the 50%/25% second-procedure ladder is mechanical: pkg + 50% + 25%xn). 5/35 bills were combos; ENT ones survive via cohort history, urology ones do not (-68%).

Method notes & caveats

- Bill basis: **net (post-concession) minus F&B**. Cash bills carry discretionary concessions (ABDUL ₹50,300; MEENAKSHI's MOU discount ₹15,954) — engine-vs-gross would shift cash errors ~+10-18% upward; verdicts were checked against both and only ABDUL/ARTH sit near the boundary.
- Extraction: gemini-2.5-flash, temp 0; 4 documents hand-verified page-by-page (ABDUL FC+FB, HEMALATHA FC, MEENAKSHI FB — all exact). One known extraction inconsistency flagged (KRANTHI FB).
- Engine controls per case: opted room from the FC (remarks wording wins), planned LOS via `los_basis: Manual`, care type from the FC header, robotic only when the FC says robotic; coblation was tracked as an equipment flag (2 ENT cases; bills carry COBLATOR ₹16,000 + PROCISE wand ₹14,687 — present in the ENT cohort history, no crash, no explicit engine line).
- SURLA was re-run after fixing a harness-side robotic regex; the residual +230k trigger is the engine's own (`treatment_text_robotic`), verified by a controlled `robotic:"no"` rebuild.
- No engine/HO source was modified; nothing committed.

Artifacts (all reproducible)

File	Content
/private/tmp/fc-eval/fc_cases.jsonl	35 extracted FC counsellings
/private/tmp/fc-eval/fb_cases.jsonl	35 matched extracted final bills
/private/tmp/fc-eval/engine_runs.jsonl	per-case flow2 + build (planned-LOS, default, replay)
/private/tmp/fc-eval/comparison.jsonl	machine three-way comparison
/private/tmp/fc-eval/adjudicated.json	final hand-adjudicated verdicts used above
/private/tmp/fc-eval/extract.mjs, run_engine.mjs, compare.py	re-runnable pipeline (skip-done)

Copies of the JSONL artifacts are alongside this report in `todo_and_helpers/engine-vs-reality-16jul-data/` .